

Athulya Ratnayake

Research Assistant • [Multidisciplinary AI Research Center \(MARC\)](#)
• University of Peradeniya, Sri Lanka

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PERSONAL DATA

Name in Full Ratnayake Mudiyansele Athulya Methsisi Bandara Ratnayake

RESEARCH INTEREST

My research interests lie at the intersection of applied mathematics, signal processing, and machine learning, with particular focus on implicit neural representations, remote sensing, random signal processing, probability theory, and optimization for developing mathematically grounded solutions to real-world signal and image analysis.

HIGHLIGHTS

- Actively contributed to a \$700k **NVIDIA-funded Heritage AI project**, developing INR-based denoising and generative AI methods to restore and enhance culturally significant historical imagery.
- **Earned national-level recognition** through the Manamperi Award and **university-level distinction** through the W.P. Jayasekara Award for the best student project in Electrical and Electronic Engineering student research project.
- **Placed in the top 100 globally** for two consecutive years in IEEEExtreme (17.0, 18.0).
- Published two Q1 journal papers and multiple peer-reviewed conference papers, with ongoing research contributions in machine learning, remote sensing, and signal processing.

EDUCATION

B.Sc. Eng. (Hons) Electrical and Electronic Engineering (First Class) June 2021 - August 2025

- University of Peradeniya, Sri Lanka
- CGPA: 3.85/4.00
- The complete degree program was conducted and assessed in English medium and accredited by Washington Accord.

SELECTED RESEARCH PUBLICATIONS

Journals

- [J1] **Mamba-FCS: Joint Spatio-Frequency Feature Fusion, Change-Guided Attention, and SeK Loss for Enhanced Semantic Change Detection in Remote Sensing**
IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing (IF: 5.3)
Buddhi Wijenayake, **Athulya Ratnayake**, Praveen Sumanasekara, Roshan Godaliyadda, Parakrama Ekanayake, Vijitha Herath, Nichula Wasalathilaka
DOI: [10.1109/JSTARS.2026.3663066](https://doi.org/10.1109/JSTARS.2026.3663066)
- [J2] **Preprocessing Algorithm Leveraging Geometric Modeling for Scale Correction in Hyperspectral Images for Improved Unmixing Performance**
IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing (IF: 5.3)
Praveen Sumanasekara, **Athulya Ratnayake**, Buddhi Wijenayake, Keshawa Ratnayake, Roshan Godaliyadda, Parakrama Ekanayake, Vijitha Herath
DOI: [10.1109/JSTARS.2026.3687834](https://doi.org/10.1109/JSTARS.2026.3687834)

- [J3] **Devising PoPStat: A Metric to Assess Mortality Dynamics within Demographic Transition**
Under Review at IEEE Access
 Tharaka Fonseka, Buddhi Wijenayake, **Athulya Ratnayake**, Inosha Alwis, Supun Sudaraka Manathunga, Roshan Godaliyadda, Samath Dharmaratne, Vijitha Herath, Parakrama Ekanayake, Isuru Thilakasiri
 arXiv DOI: [10.48550/arXiv.2501.11514](https://arxiv.org/abs/10.48550/arXiv.2501.11514)
- [J4] **Unsupervised learning reveals global demographic configurations and national trajectories from population pyramids**
Under Review at Nature Humanities and Social Sciences Communications
Athulya Ratnayake, Lelumi Edirisinghe, Thisumi Wijesinghe, Roshan Godaliyadda, Parakrama Ekanayake, Vijitha Herath, Samath Dharmaratne

Conferences

- [C1] **Enhanced SCanNet with CBAM and Dice Loss for Semantic Change Detection**
 2025 Moratuwa Engineering Research Conference (MERCon)
Athulya Ratnayake, Buddhi Wijenayake, Praveen Sumanasekara, Roshan Godaliyadda, Vijitha Herath, Parakrama Ekanayake
 DOI: [10.1109/MERCon67903.2025.11217111](https://doi.org/10.1109/MERCon67903.2025.11217111)
- [C2] **Precision Spatio-Temporal Feature Fusion for Robust Remote Sensing Change Detection**
 IEEE 19th International Conference on Industrial and Information Systems (ICIIS) 2025
 Buddhi Wijenayake, **Athulya Ratnayake**, Praveen Sumanasekara, Nichula Wasalathilaka, Mathivathanan Piratheepan, Roshan Godaliyadda, Mervyn Ekanayake, Vijitha Herath
 DOI: [10.1109/ICIIS69028.2026.11450773](https://doi.org/10.1109/ICIIS69028.2026.11450773)
- [C3] **PoPStat-COVID19: Leveraging Population Pyramids to Quantify Demographic Vulnerability to COVID-19**
 25th International Conference on Advances in ICT for Emerging Regions
 Buddhi Wijenayake, **Athulya Ratnayake**, Lelumi Edirisinghe, Uditha Wijeratne, Tharaka Fonseka, Roshan Godaliyadda, Samath Dharmaratne, Parakrama Ekanayake, Vijitha Herath, Inosha Alwis, Supun Manathunga
 arXiv DOI: [10.48550/arXiv.2509.14213](https://arxiv.org/abs/10.48550/arXiv.2509.14213)
- [C4] **ScaLe-INR: Scale and Learn Implicit Neural Representations**
 Under Review at International Conference on Learning Representations (ICLR) 2027
 Buwaneka Epakanda, **Athulya Ratnayake**, Pandula Thennakoon, Mario De Silva, Avishka Ranasinghe, Roshan Godaliyadda, Parakrama Ekanayake
 arXiv DOI: <https://doi.org/10.48550/arXiv.2606.27862>
- [C5] **From Parallelism to Specialisation: SSA-Guided Multi-Branch Implicit Neural Representations**
 Under Review at International Conference on Learning Representations (ICLR) 2027
Athulya Ratnayake, Buwaneka Epakanda, Sajana Bogahawatta, Roshan Godaliyadda, Parakrama Ekanayake

ORAL PRESENTATIONS

- **Correlation Analysis of Age Structure Metrics and Cause-Specific Mortality: A Global and WHO Regional Comparison Using the Top 5 Causes of Death in 2021**
 1st International Conference on Applied Sciences - 2025, Sri Lanka
[Presentation](#) | [Certificate](#)
- **Enhanced SCanNet with CBAM and Dice Loss for Semantic Change Detection**
 2025 Moratuwa Engineering Research Conference (MERCon)
[Presentation](#)
- **Precision Spatio-Temporal Feature Fusion for Robust Remote Sensing Change Detection**
 IEEE 19th International Conference on Industrial and Information Systems (ICIIS) 2025

SELECTED PROJECTS

Satellite Image-Based Remote Sensing for Change Detection and Landscape Mapping

September 2024 - August 2025

Undergraduate Final Year Project

[Project Report](#)

- Conducted a remote sensing research project covering semantic change detection and hyperspectral unmixing for improved landscape analysis.
- Developed a spatio-frequency fusion and change-guided attention framework to improve boundary preservation, class imbalance handling, and fine-scale change detection.
- Integrated CBAM and task-specific loss functions to refine decoder representations and improve semantic segmentation quality.
- Proposed a perspective-projection-based scale-correction method for hyperspectral unmixing, restoring Linear Mixing Model geometry under illumination-induced distortions.
- Reduced abundance estimation errors by approximately 50% across diverse state-of-the-art unmixing methods, demonstrating the method as a plug-and-play preprocessing enhancement.

Implicit Neural Representations: Coordinate Scaling and Branch Specialisation

September 2025 - Present

Present

Ongoing

- Developed CoRe-INR using axis-wise coordinate scaling and confidence-based fusion; analysed spectral support and convergence through frequency-domain and neural tangent kernel analysis.
- Proposed SSA-guided branch specialisation with explicit component supervision, PSO-initialised learnable scaling, and gated additive fusion for image and audio representation.
- Evaluated CoRe-INR on image, audio, and 3D representation, alongside denoising, super-resolution, and inpainting.
- Contributing to HeritageAI, an NVIDIA-funded MARC subproject focused on restoring historical photographs, documents, and cultural imagery.

A Data-Driven Demographic Analysis Approach for Understanding Mortality

Variation Across Nations

June 2023 - Present

Ongoing

- Developed PoPDivergence, a KL-divergence-based scalar metric that summarizes full population-pyramid structures for cross-country mortality analysis.
- Introduced PoPStat to quantify the relationship between demographic structure and disease-specific mortality across 371 global diseases.
- Designed a reference-tuning algorithm to identify demographic structures that maximize mortality correlation.
- Benchmarked PoPStat against HDI, GDP, median age, and life expectancy, showing stronger explanatory power for several major diseases.

AWARDS AND ACHIEVEMENTS

Manamperi Award (Engineering) | [Certificate](#)

2025

- Awarded to outstanding undergraduate Engineering Research Projects carried out at a Faculty of Engineering in a Sri Lankan University or a Higher Education Institute
- Merit Award

W.P. Jayasekara Prize | [Certificate](#)

2025

- Awarded for the Best Student Project in Electrical and Electronic Engineering, University of Peradeniya.

IEEEEXTreme

2024

- IEEEExtreme 18.0 - **Global Rank: 66, Country Rank: 01** | [Certificate](#) 2024
- IEEEExtreme 17.0 - **Global Rank: 65, Country Rank: 01** | [Certificate](#) 2023
- Coders - The largest inter-university algorithmic coding competition in Sri Lanka** 2024
- Coders v11.0 - Champions | [Certificate](#) 2024
- Coders v10.0 - 2nd runners up 2023
- Coders v9.0 - Champions | [Certificate](#) 2022
- Other Competitive Programing Competitions and Hackathons** 2024
- 2nd Runners-Up at RealHack 5.0 Islandwide Hackathon | [Certificate](#) 2024
- 2nd Runners-Up at Cypher 2.0 All Island Competitive Programming Competition | [Certificate](#) 2024
- Champions at MoraXtreme 8.0 All Island Competitive Programming Competition 2023

WORK EXPERIENCE

AI Research and Development Engineer August 2026 - Present
[Farbe Technologies \(Pvt\) Ltd](#)

- Working as an AI Research and Development Engineer, providing AI-based solutions to practical problems in industry.
 - Conducting applied research in artificial intelligence and machine learning.
 - Designing, developing, and evaluating AI models tailored to industrial requirements.
 - Translating research into practical tools for automation, process optimisation, and data-driven decision-making.

Research Assistant (Volunteer) February 2023 - Present
[Multidisciplinary AI Research Center \(MARC\)](#), University of Peradeniya

- Conducting research in implicit neural representations (INRs), remote sensing, and demographic analysis.
- Conducting AI awareness sessions for the general public to build understanding of artificial intelligence and its practical applications.

Research Assistant August 2025 - June 2026
[Multidisciplinary AI Research Center \(MARC\)](#), University of Peradeniya

- Worked as a Research Assistant in the field of AI for Social Sciences and Policies.
- Conducting research on,
 - Implicit Neural Representations (INRs)
 - Demographic analysis for data-driven policy making
- Conducted research in the field of Remote Sensing including Change Detection and Hyperspectral Unmixing

Casual Instructor (Casual Teaching Assistant) March 2024 - June 2024
 Department of Computer Engineering, Faculty of Engineering, University of Peradeniya

- Served as a casual instructor for the course CO253 – Introduction to Programming and Networking for Electrical Engineering.
- Assisted students in understanding theoretical concepts and provided hands-on guidance during laboratory sessions.

Intern Research Assistant July 2023 - October 2023
 National Institute of Fundamental Studies (NIFS), Kandy

- Worked in the Material Processing and Device Fabrication Laboratory, focusing on supercapacitor technology.
- Analyzed the performance of nitrogen-doped activated carbon supercapacitors and optimized materials, achieving a 15% increase in capacitance.

EXTRA-CURRICULAR

Clubs and Societies

- Founder President of the Engineering Mathematics Society, University of Peradeniya 2024-2025
- Foundation of Astronomical Studies and Exploration (FASE)
 - Treasurer (2023), Chief Organizer (2022), Vice President (2021)

Volunteer Experiences

- MARC AI Outreach Programs 2024-Present
 - Voluntarily organized, compèred, and delivered AI outreach sessions across Sri Lanka to demystify AI among school and university students, encouraging them to become creators of AI rather than mere consumers.
 - Conducted programs across both rural and urban regions of Sri Lanka including sessions for University of Vavuniya, South Eastern University of Sri Lanka, and pre-engineering students across Sri Lanka.
- Delivered an invited session on IEEEExtreme strategies at “Winning Strategy – Learn from Sri Lanka’s No.1”, organized by the IEEE Computer Society of SLIIT, highlighting competitive programming, problem-solving, and team coordination. 2025
- Guest speaker at “CODEX Session Series”, organized by the IEEE Sri Lanka Section, delivering a talk on Dynamic Programming. — [Session Recording](#) 2025
- Guest speaker at “ALGOFORGE”, organized by the IEEE Sri Lanka Section, delivering a talk on Number Theory and Combinatorics. — [Session Recording](#) 2024
- Guest speaker at “Path to Xtreme”, organized by the IEEE Sri Lanka Section, conducting an awareness session on IEEEExtreme. — [Session Recording](#) 2024

REFERENCES

Prof. G.M. Roshan I. Godaliyadda
Ph.D.(National University of Singapore)

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